Module 1

1. What is Software?

Software is a set of instructions, data, or programs used to operate computers and execute specific tasks. It's the intangible part of a computer system, contrasting with hardware, which is the physical component. Software includes everything from operating systems and applications to programming tools and libraries.

2. What is Software Engineering?

Software engineering is a systematic, disciplined, and quantifiable approach to the development, operation, and maintenance of software. It applies engineering principles to software creation to ensure efficiency, reliability, maintainability, and quality, especially for large and complex systems.

3. Explain Types of Software

Software can be broadly categorized into several types:

System Software: This software manages and controls the computer hardware and provides a platform for other software to run. Examples include Operating

Systems (Windows, macOS, Linux), Device Drivers, and Utilities.

Application Software: These are programs designed to perform specific tasks for the end-user. Examples include word processors (Microsoft Word), web browsers (Chrome, Firefox), media players, accounting software, and mobile apps.

Programming Software (Development Tools): These are tools that assist programmers in writing, testing, and debugging other software. Examples include Integrated Development Environments (IDEs like VS Code, PyCharm), compilers, debuggers, and text editors.

Embedded Software: Software embedded into hardware devices to control their functions. Examples include software in washing machines, smart TVs, car systems, and medical equipment.

Web Applications: Software accessed via a web browser over the internet, often without needing to be installed directly on the user's device. Examples include online banking, social media platforms (Facebook, Instagram), and e-commerce sites (Flipkart).

4. What is SDLC? Explain Each Phase of SDLC

SDLC stands for Software Development Life Cycle. It is a structured process that enables the production of high-quality, low-cost software in the shortest possible production time. It's a framework defining tasks performed at each step in the software development process.

The typical phases of SDLC are:

Requirement Gathering and Analysis: This initial phase involves collecting detailed requirements from the client or stakeholders. It defines what the software needs to do, including functional and non-functional requirements. This is usually done through interviews, surveys, and analysis of existing systems.

Design: In this phase, the gathered requirements are transformed into a design plan. This includes architectural design (overall structure), user interface (UI) design, database design, and algorithm design. The goal is to define how the software will meet the requirements.

Implementation/Coding: This is where the actual code is written by developers based on the design

specifications. Programmers use suitable programming languages and development tools to build the software components.

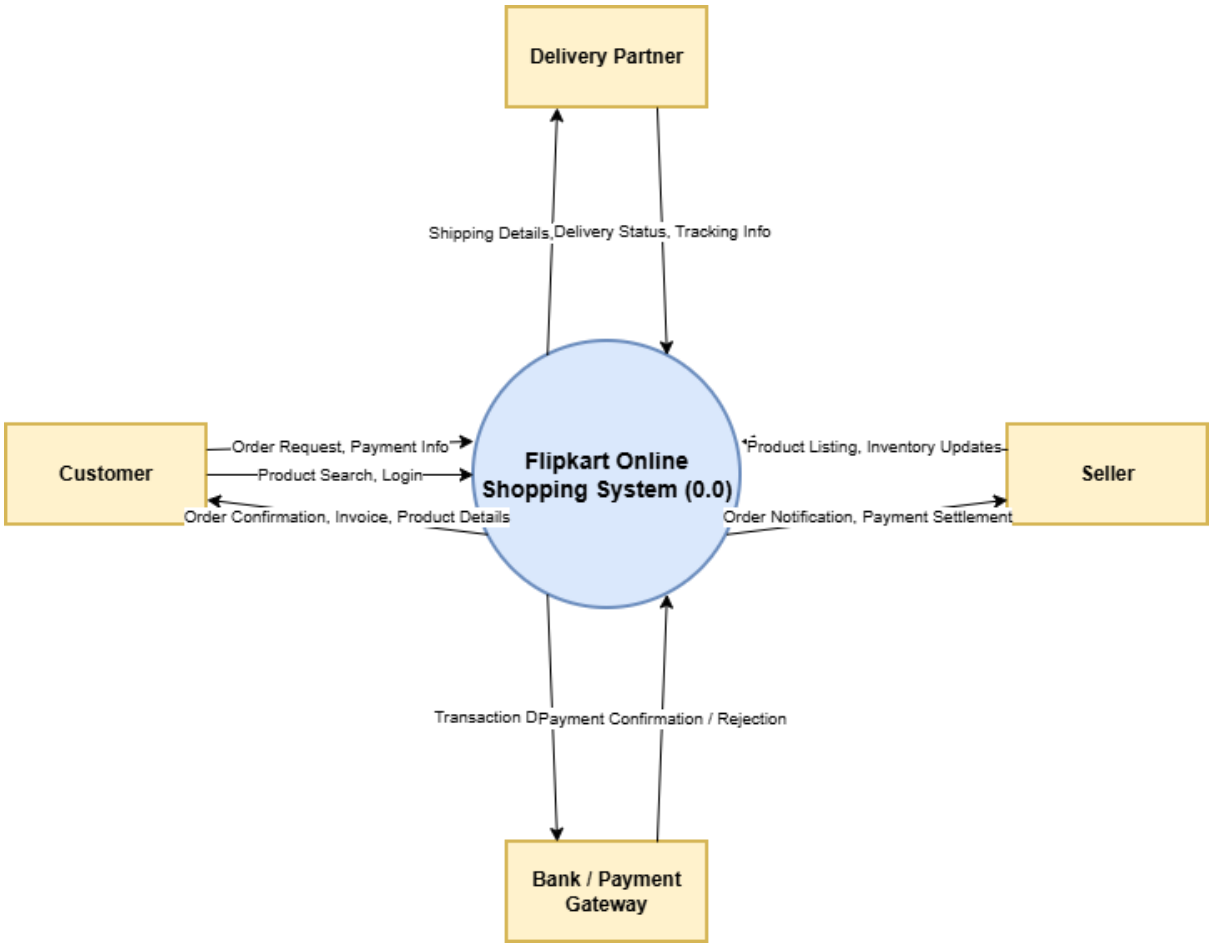
Testing: After implementation, the software is rigorously tested to identify and fix defects, bugs, and errors. Various testing types are performed, such as unit testing, integration testing, system testing, and user acceptance testing (UAT), to ensure it meets the specified requirements.

Deployment: Once the software passes all testing phases and is approved, it is released to the end-users. This can involve installing the software on servers, making it available through app stores, or deploying it on cloud platforms.

Maintenance: This is the ongoing phase after deployment. It involves monitoring the system for issues, providing updates, fixing new bugs that arise, enhancing features, and adapting the software to changes in the environment or user needs.

5. What is DFD? Create a DFD Diagram on Flipkart

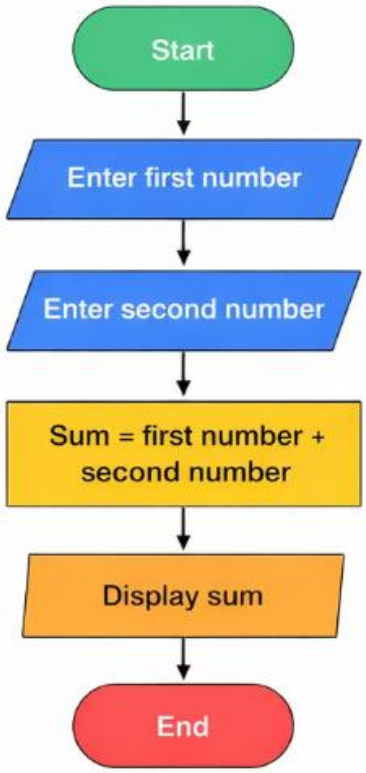
DFD stands for Data Flow Diagram. It is a graphical representation of the "flow" of data through an information system. A DFD shows how information enters and leaves the system, what changes the information, and where the information is stored. It doesn't show the process logic, only the data paths.



6. What is Flowchart? Create a Flowchart to Make Addition of Two Numbers

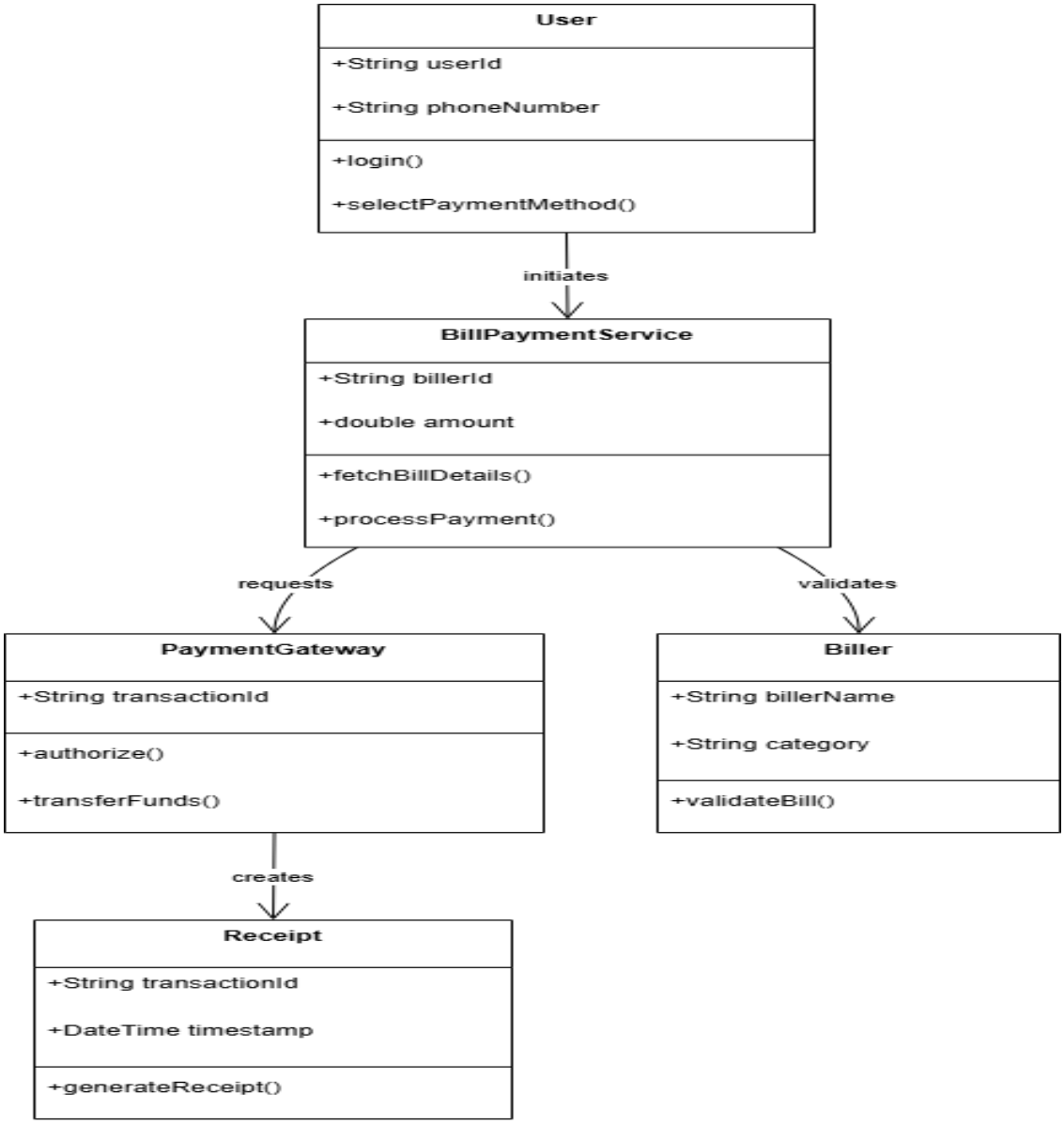
A Flowchart is a diagrammatic representation of an algorithm, workflow, or process. It uses standardized graphical symbols to depict the steps and decision points in a sequence, illustrating the flow of control from one point to another.

Flowchart to Make Addition of Two Numbers:



7. What is Use Case Diagram? Create a Use-Case on Bill Payment on Paytm.

A Use Case Diagram is a graphical representation of the functional requirements of a system**.** It shows how different users (actors) interact with the system to achieve specific goals (use cases). It describes *what* the system does from the perspective of external users, without detailing how it does it.



Description of Elements:

System Boundary (Rectangle): Represents the scope of the system (e.g., "Paytm System"). Use cases are inside, actors are outside.

Actor (Stick Figure): Represents a role played by a user or another system that interacts with the system. (e.g., "User", "Payment Gateway")

Use Case (Oval): Represents a distinct functional goal achieved by the actor using the system. (e.g., "Pay Bill", "View Bill History", "Select Payment Method", "Process Payment")

Association (Line): Connects actors with the use cases they participate in. An arrow indicates the direction of initiation or data flow.

<<include>> (Dashed Arrow): Indicates that one use case always includes the functionality of another use case. (Implied: "Pay Bill" and "View Bill History" might include "Select Payment Method" and "Process Payment").

<<extend>> (Dashed Arrow): Indicates that one use case optionally extends the functionality of another use case under certain conditions.